

Do Not Anonymize Away the Answer: Measuring Task-Critical Meaning Loss in Text De-identification

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Abstract

Text de-identification is often evaluated as a span removal problem: did the system remove names, dates, locations, and other identifiers? For sensitive text used in foundation-model workflows, this is necessary but incomplete. A transformed clinical vignette or legal case excerpt can pass a privacy-oriented screen while removing the medication dose, lab value, statutory article, or legal claim needed for downstream reasoning. We introduce a small clinical/legal benchmark for utility-preserving anonymization, a task-critical fact retention metric, and Critical Span Guard, a low-cost protocol that extracts privacy spans and task-critical facts, anonymizes under preservation constraints, verifies the result, and applies a deterministic safety/repair layer. In a 150-example non-oracle LLM run, Critical Span Guard achieves zero direct identifier leaks, lower quasi-identifier risk (0.127 versus 1.733 for generic prompting and 0.573 for privacy-first prompting), and higher audited task-critical fact retention (0.994 versus 0.938 and 0.529). A separate handwritten stress slice shows why the problem is not reducible to “redact more”: some sensitive facts, such as pregnancy or sexual orientation, can be the task-critical answer itself. The results support evaluating de-identification systems by both privacy protection and retained domain meaning.

1 Introduction

Sensitive clinical and legal text is valuable for retrieval, supervised fine-tuning, evaluation, auditing, and synthetic-data generation. The usual first step before such use is de-identification or anonymization. However, a system optimized only for removing personally identifying information can destroy the very facts that make the text useful. In a clinical note, replacing “methotrexate 15 mg weekly” with “a medication” may preserve broad topical similarity while breaking the answer to a treatment question. In a legal excerpt, replacing “Article 6” with “a specified article” removes the legal basis needed to reason about the claim.

This paper studies the question: after anonymization, is the transformed text still useful for the domain task it was meant to support? We focus on two domains where this tension is concrete. Clinical-style vignettes contain diagnoses, medications, doses, labs, symptoms, and timelines. Public legal snippets derived from the Text Anonymization Benchmark (TAB) [Pilan et al. \(2022\)](#) contain application identifiers, party roles, claims, legal articles, procedural facts, and outcomes. Existing tooling such as Microsoft Presidio [Microsoft \(2026\)](#) and regulatory frameworks such as HIPAA de-identification guidance [U.S. Department of Health and Human Services \(2012\)](#) are essential privacy infrastructure, but they do not by themselves answer whether the remaining text preserves task-critical meaning.

We make three contributions:

1. A 150-example controlled benchmark with synthetic clinical vignettes and TAB-derived legal snippets, each annotated with direct identifiers, quasi-identifiers, task-critical facts, and downstream QA items.

2. *Task-Critical Fact Retention* (TCFR), a fact-level utility metric that distinguishes retained domain meaning from generic surface similarity.
3. *Critical Span Guard*, a simple anonymization protocol that extracts privacy spans and task-critical facts, anonymizes under explicit preservation constraints, and applies a deterministic safety/repair layer for direct identifiers and known domain-specific failure modes.

Our goal is not to claim formal anonymization, compliance, or broad clinical deployment. Instead, we show that span-level privacy evaluation alone can both miss leaks and reward over-redaction, and that small task-aware checks expose a more useful privacy–utility frontier.

Positioning. Clinical de-identification research has produced strong PHI-detection systems and shared-task style evaluations for patient notes [Dernoncourt et al. \(2016\)](#), while TAB extends anonymization evaluation to public legal text with entity, masking, confidentiality, and coreference annotations [Pilan et al. \(2022\)](#). Recent anonymization benchmarks increasingly evaluate privacy and utility together [Loiseau et al. \(2025\)](#); [Krčo et al. \(2026\)](#). Our contribution is orthogonal: we make the utility target explicit at the level of domain facts and downstream QA, then show that generic text overlap and span removal can miss clinically or legally material failures.

2 Benchmark

Clinical split. We generate 75 synthetic clinical vignettes from templates. Each example contains direct identifiers such as patient name, clinician name, clinic, date, medical record number, phone number, and email; quasi-identifiers such as age, city, and occupation; and task-critical facts such as diagnosis, medication, dose, lab result, symptom duration, imaging result, or treatment plan. The clinical split is synthetic by design: it provides controlled utility labels without exposing real protected health information.

Legal split. We sample 75 short excerpts from TAB, an open-source corpus of English-language European Court of Human Rights cases annotated with personal identifiers, masking decisions, confidential attributes, and coreference relations [Pilan et al. \(2022\)](#). For each sampled snippet, we keep short procedure, complaint, and outcome sentences when available. We use TAB spans for privacy labels and heuristic extraction plus manual spot checks for legal articles, claims, outcomes, and QA answers.

Stress slice. We also construct a 12-example handwritten diagnostic slice that stresses cases where privacy and utility overlap: pregnancy, sexual orientation, domestic violence, Roma marriage, exact ages, rare occupations, case labels, secret evidence, exact legal article/protocol specificity, numeric labs, and dose preservation. This slice is not used as the headline non-oracle result; it is an interpretability check on the privacy–utility frontier.

3 Metrics

We report privacy and utility metrics together. *Direct leak* is the fraction of rows where any gold direct identifier remains in the anonymized text. *QI risk* is the mean quasi-identifier risk score, where specific dates, locations, occupations, sensitive attributes, or combinations of details increase the risk. We also audit direct, quasi, and all-span privacy recall: the fraction of gold privacy spans not found verbatim after anonymization. *Exact TCFR* is the fraction of gold task-critical facts found by deterministic phrase or alias matching. *Audited TCFR* uses a structured LLM fact judge to distinguish faithful paraphrase from real omission. *QA consistency* asks whether downstream questions still recover the gold answers from the anonymized text.

Exact matching is reproducible but conservative. Audited TCFR is more semantically aware but should not be treated as ground truth. We therefore use manual spot checks for foreground examples and explicitly mark where exact and audited metrics disagree.

Method	Direct leak ↓	QI risk ↓	Exact TCFR ↑	Audited TCFR ↑	QA ↑
Generic LLM	0.180	1.733	0.813	0.938	0.888
Privacy-first LLM	0.000	0.573	0.398	0.529	0.414
Critical Span Guard	0.000	0.127	0.884	0.994	0.974

Table 1: Main 150-example non-oracle results.

Stage	Direct leak	QI risk	Exact TCFR	QA
CSG draft	0.073	1.027	0.881	0.969
CSG verified	0.073	1.027	0.881	0.969
CSG + safety/repair	0.000	0.127	0.884	0.974

Table 2: Critical Span Guard ablation reconstructed from cached outputs.

4 Methods

We compare three non-oracle LLM anonymizers in the 150-example OpenAI run. The *generic LLM* prompt asks the model to remove personally identifying information while preserving meaning. The *privacy-first LLM* prompt asks the model to remove or generalize any information that could identify a person, including rare locations, exact dates, named institutions, rare occupations, and unique events.

Critical Span Guard uses the same low-cost model but adds structure. First, it extracts privacy spans and task-critical facts from the input. Second, it anonymizes using those structures, with rules that preserve clinical medications, doses, labs, diagnoses, symptoms, timelines, and legal article/statute references, claims, procedures, outcomes, and reasoning unless they are direct identifiers. Third, it runs a verification/repair prompt. Finally, it applies a deterministic safety/repair layer that replaces direct identifiers such as case/application numbers, dates, MRNs, phone numbers, emails, names, and institutions, and repairs known domain mistakes such as legal Article over-generalization and medication placeholdering.

We also run local diagnostic baselines on the original benchmark, a 200-example local expansion, and the stress slice: regex rules, Presidio, direct-span oracle replacement, privacy-first oracle redaction, and a CSG oracle using gold benchmark annotations. These oracle variants are not deployable systems; they diagnose upper bounds and failure modes.

5 Results

Table 1 shows the core privacy–utility tradeoff. Generic prompting preserves many task facts, but leaks direct identifiers in 27 of 150 examples and has high QI risk. Privacy-first prompting removes all measured direct identifiers, but exact TCFR falls to 0.398 and QA consistency to 0.414. Critical Span Guard achieves zero direct leaks, the lowest overall QI risk, audited TCFR of 0.994, and QA consistency of 0.974. The span-level privacy audit agrees: Critical Span Guard reaches direct span recall of 1.000 and all privacy-span recall of 0.982, compared with 0.919 and 0.771 for generic prompting and 1.000 and 0.964 for privacy-first prompting. Paired deltas on the same examples sharpen this interpretation: relative to generic prompting, Critical Span Guard reduces direct-leak rate by 0.180 (95% bootstrap CI 0.120–0.247) and QI risk by 1.607 (1.407–1.807); relative to privacy-first prompting, it improves exact TCFR by 0.487 (0.434–0.538), audited TCFR by 0.466 (0.416–0.514), and QA consistency by 0.560 (0.498–0.620). A 200-example local/oracle diagnostic expansion gives the same qualitative frontier: regex and direct-span replacement preserve exact TCFR but leave high QI risk (1.780 and 2.810), Presidio misses many legal identifiers (0.980 direct-leak rate on legal examples) and loses clinical facts (0.445 clinical TCFR), and privacy-first oracle redaction protects privacy at large utility cost (0.448 overall TCFR). Figure 1 visualizes the same non-oracle tradeoff using QI risk and audited TCFR.

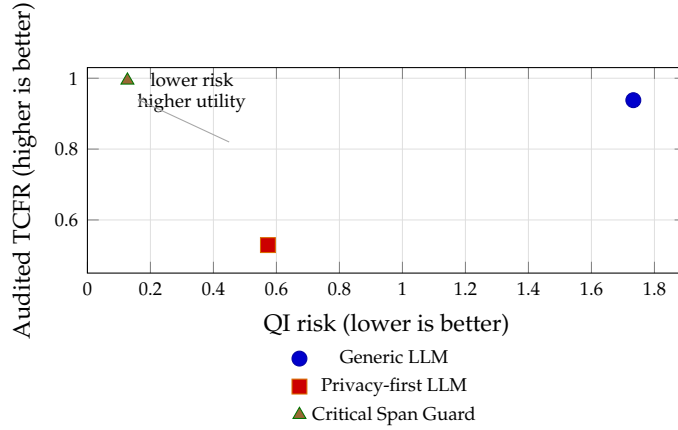


Figure 1: Privacy-utility frontier for the 150-example non-oracle LLM run. Critical Span Guard is closest to the upper-left target region: lower measured quasi-identifier risk and higher audited task-critical fact retention.

Method	Token F1	TF-IDF	Edit rate	Audited fact-fail rate	Privacy-fail rate
Generic LLM	0.776	0.708	0.247	0.240	0.747
Privacy-first LLM	0.577	0.472	0.464	0.793	0.273
Critical Span Guard	0.773	0.617	0.274	0.020	0.027

Table 3: Surface similarity metrics are not sufficient proxies for privacy-safe utility.

128 The ablation in Table 2 is important for claim discipline. In this cached sample, the verification/repair prompt did not change deterministic metrics: the draft and verified rows are
 129 numerically identical. The measured gain comes from the deterministic safety/repair layer,
 130 which closes eleven direct leaks, reduces QI-risk rows from 74 to 11, repairs legal Article
 131 specificity losses, and leaves eleven legal residual cases.
 132

133 Table 3 addresses whether generic utility metrics are enough. Generic prompting and
 134 Critical Span Guard have nearly identical token F1 against the original text, and generic
 135 prompting has higher TF-IDF cosine similarity, but it also has far higher privacy failure and
 136 audited fact failure rates. Among the 225 method-example rows at or above the median
 137 TF-IDF cosine score, 73 still have privacy failures and 34 still have audited fact loss. Surface
 138 similarity therefore rewards text that may be too close to the original and misses task-critical
 139 failures; this is consistent with broader concerns that generic text-generation metrics do not
 140 encode domain-specific faithfulness by themselves Zhang et al. (2020). Figure 2 shows the
 141 same failure mode at the row level.

142 The stress slice in Table 4 clarifies the frontier. Privacy-first oracle redaction drives measured
 143 direct leaks and QI risk to zero, but fails QA on 11 of 12 rows. Direct-span replacement
 144 preserves utility but leaves quasi-identifiers in every row. CSG oracle replacement preserves
 145 all task facts with zero direct leaks, but retains task-critical quasi-identifiers in 6 of 12 rows.
 146 This residual risk is not merely an implementation bug: some facts, such as pregnancy,
 147 sexual orientation, domestic violence, or minority status, can be both sensitive and necessary
 148 to answer the domain question.

149 6 Qualitative Examples

150 In `clinical_0006`, the original vignette describes rheumatoid arthritis with MCP joint
 151 swelling, positive anti-CCP, and methotrexate 15 mg weekly plus folic acid 1 mg daily. The
 152 privacy-first output replaces these with “finger joint areas,” “a specific antibody test,” “an
 153 autoimmune joint condition,” and “a disease-modifying medication,” so both clinical QA

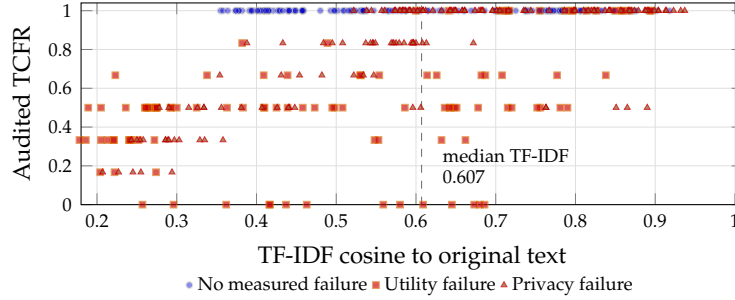


Figure 2: Surface similarity failure scatter for the 150-example non-oracle run. Among the 225 rows at or above the median TF-IDF cosine, 73 still have privacy failures and 34 have audited fact loss.

Method	Direct leak	QI risk	Exact TCFR	QA
Regex	0.083	2.167	1.000	1.000
Presidio	0.667	2.667	0.817	0.833
Direct-span oracle	0.000	2.500	1.000	1.000
Privacy-first oracle	0.000	0.000	0.608	0.528
CSG oracle	0.000	0.667	1.000	1.000

Table 4: Handwritten robustness stress slice. This diagnostic slice is separate from the non-oracle headline run.

154 questions fail. Critical Span Guard redacts the patient, clinic, date, MRN, phone, email,
155 clinician, city, and exact age while preserving the lab, diagnosis, medication, and doses.

156 In legal_0006, generic prompting leaves application number 37645/97 and the exact
157 filing date while preserving Article 6. The privacy-first output removes those identifiers but
158 replaces Article 6 with “a specified article,” breaking the legal-basis question. Critical Span
159 Guard removes the application number, name, country, institution, nationality, and date
160 while preserving Article 6 and the reasonable-time civil-proceedings claim.

161 The remaining frontier is visible in legal_0005: generic and privacy-first prompting avoid
162 the measured QI hits by generalizing away the specific section 42 Fair Employment claim,
163 while Critical Span Guard preserves Article 6 and the exact claim. The preserved statute
164 name still contains “Northern Ireland” and “1976,” so the row remains a measured QI-risk
165 case. This is a useful limitation: some legal statute, jurisdiction, regime, or offense specificity
166 may be both important for legal reasoning and quasi-identifying. A residual audit of the
167 eleven remaining CSG QI rows found six clear task-critical legal-overlap cases and five
168 additional frontier candidates.

169 7 Limitations and Ethics

170 This is a small fast-iteration study, not a formal anonymization guarantee. The clinical split
171 is synthetic and should not be presented as evidence on real clinical notes. The legal split
172 is public TAB-derived text, but legal fact/claim extraction is heuristic and manually spot-
173 checked rather than fully expert annotated. Audited TCFR uses an LLM judge; it is useful for
174 separating paraphrase from omission, but manual review remains necessary for final paper
175 claims. The privacy metrics detect known direct spans and heuristic quasi-identifiers; they
176 do not provide k-anonymity, differential privacy, legal compliance, or protection against a
177 motivated adversary with external knowledge. The remaining CSG quasi-identifier rows
178 should be read as audit targets, not as solved cases: several involve quasi-identifying text
179 inside the annotated legal claim, and five are frontier candidates requiring policy review. A
180 more privacy-aggressive release policy could generalize those claims, but that would change
181 the downstream legal utility target. In the promoted 150-example run, three CSG rows

have audited fact loss under strict labeling: `legal_0016` generalizes “Polish authorities” to “authorities,” while `clinical_0055` and `clinical_0059` generalize or omit strict clinical specificity. These rows mark jurisdictional, temporal, and diagnostic specificity as policy-sensitive utility targets rather than privacy failures.

Threat model. We evaluate direct string leakage and heuristic quasi-identifier retention under a bounded controlled-release scenario: an inspector can search transformed text for known direct identifiers and retained quasi-identifying details. We do not model linkage attacks by an adversary with external databases or case-specific background knowledge. Public redistribution should therefore preserve provenance and license information and include residual-risk notes, especially for TAB-derived legal snippets.

The intended use of this work is evaluation and safer data transformation research. We do not release or process real protected health information. The main ethical lesson is that aggressive redaction can make sensitive data safer to share only by making it useless or misleading. Responsible release pipelines should therefore state the downstream utility requirements explicitly and audit both residual privacy risk and task-critical meaning loss.

8 Conclusion

De-identification systems should be evaluated not only by what they remove, but also by whether they preserve the facts needed for downstream reasoning. On a controlled clinical/legal benchmark, generic anonymization leaks identifiers and quasi-identifiers, privacy-first anonymization often removes the answer, and a small amount of task-aware structure improves the observed privacy–utility tradeoff. The stress slice further shows that some privacy–utility conflicts are inherent: the sensitive fact may also be the task-critical fact. This makes task-critical utility evaluation a necessary complement to span-level privacy metrics for sensitive text transformation.

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